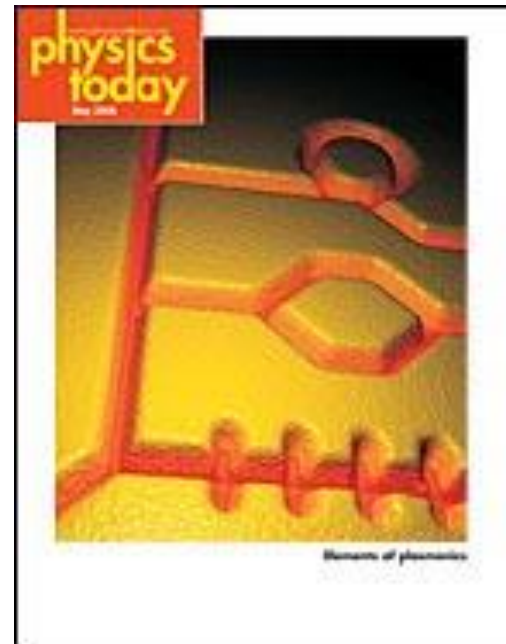


Plasmonic materials and devices

- Circuit model
- Drude model (revision)
- **Localized Surface Plasmons**
- Surface Plasmon Polaritons



Plasmons

Plasma = gas of free charge carriers, e.g. the carriers in a conductor

‘on’ = wavicle

A surface plasmon is a

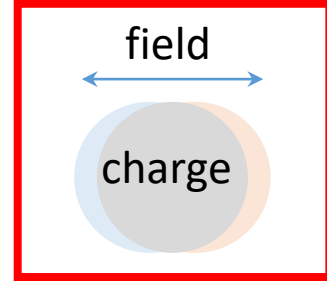
- collective oscillation of charge carriers near a surface
- a quasiparticle hybrid between electron and photon
- an evanescent wave on a conducting surface...

There are two limiting cases of plasmons

- ❖ LSP = localized surface plasmon (e.g. on a small particle)
- ❖ SPP = surface plasmon polariton (e.g. on a metal film)

The distinction can become less obvious in-between these two cases

Localized Surface Plasmons (LSP): historical development



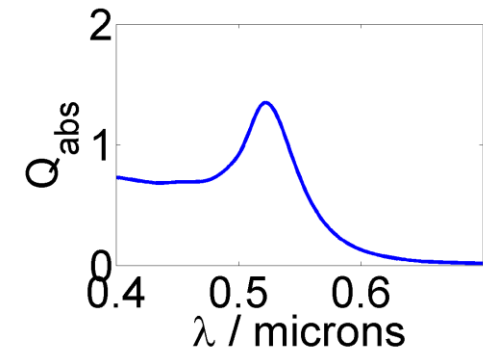
Lycurgus cup: made 4th century



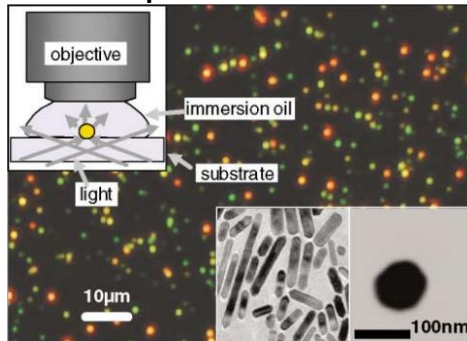
British Museum

Resonant

Optical spectroscopy:
18th & 19th centuries



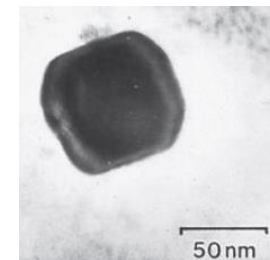
Dark-field microscopy
developed 1903



Sönnichsen et al (2002)

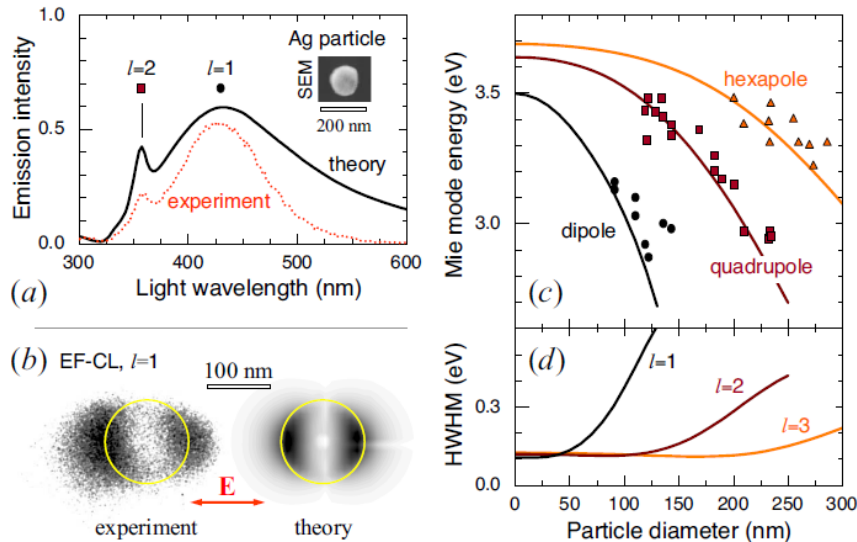
PRL **88** 077402

TEM (developed 1930s)



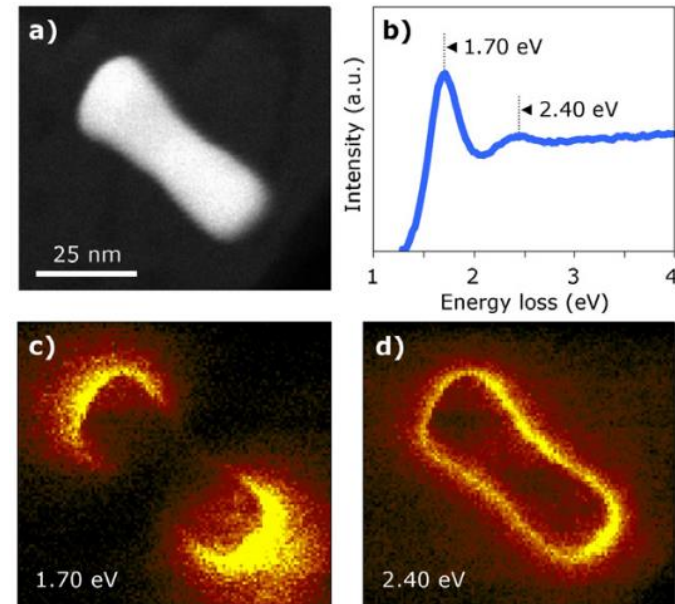
LSP nanoscale mapping

Cathodoluminescence



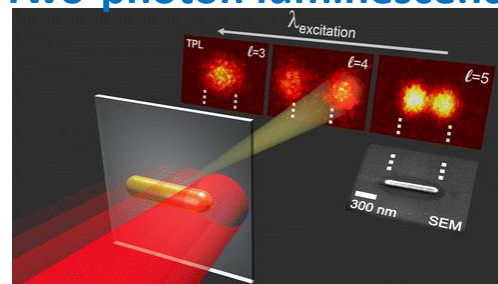
Yamamoto et al (2001), PRB **64** 205419

Electron Energy Loss

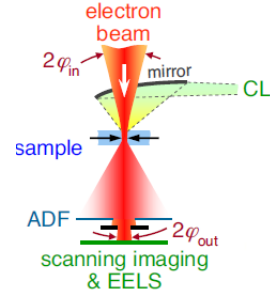


Bosman et al (2007) Nanotech **18** 165505

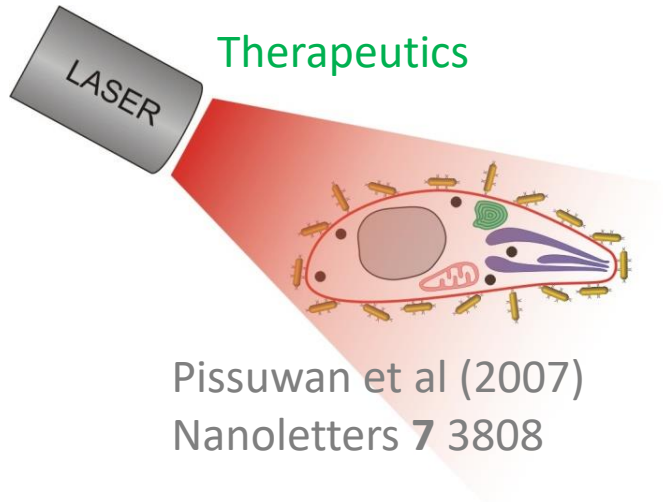
Two-photon luminescence



ACS Photonics, 2015, 2, 410

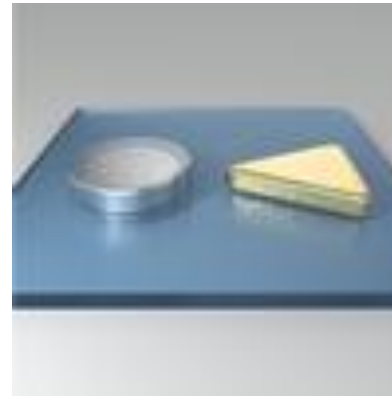


Localized Surface Plasmons: applications



1. Antibody coated particle sticks to target
2. Laser passes through body
3. Particles absorb $\rightarrow$ heat $\rightarrow$ localized kill

Sensing



[dx.doi.org/10.1364/OME.2.000111](https://doi.org/10.1364/OME.2.000111)

1. Analyte changes local refractive index
2. LSP peak shifts
3. Detected

Example above:

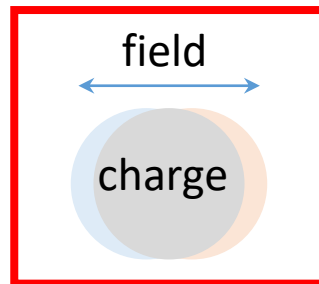
Pd disc changes n when absorbs H

Au triangle resonance shifts

Understanding LSP – charge model

(because we want to know to control frequency etc)

- Drude (free e-) model explains conductor response
- Excitation field forces opposite charge to boundaries
- Separated charge provides restoring force
- Looks like bound charge (surface) → Lorentz oscillator
- Boundary shape → distribution → force → energy



Quasistatic model

(Lorentz oscillator)

- Assumes particle is so small that
 - “sloshing” time is negligible
 - field penetrates fully through particle
 - response mainly due to internal displacement field (not currents)
- Geometry and material properties “separable”
- Maximum response at resonance ($1/\chi \approx -S$)
- **Limiting factor:** χ is complex

$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi} \equiv \frac{1}{\varepsilon / \varepsilon_b - 1}$$

Size Shape Material

A: Strength (0 to 1)
 S: Position (0 to 1)
Sphere dipole=1/3 Rod dipoles: transverse→1/2, long→0
Disc dipoles: plane→0, normal→1 $\varepsilon_b = \text{background}$

LSP: effect of metal

Position explained by factors affecting valence density

Height mostly optical resistivity

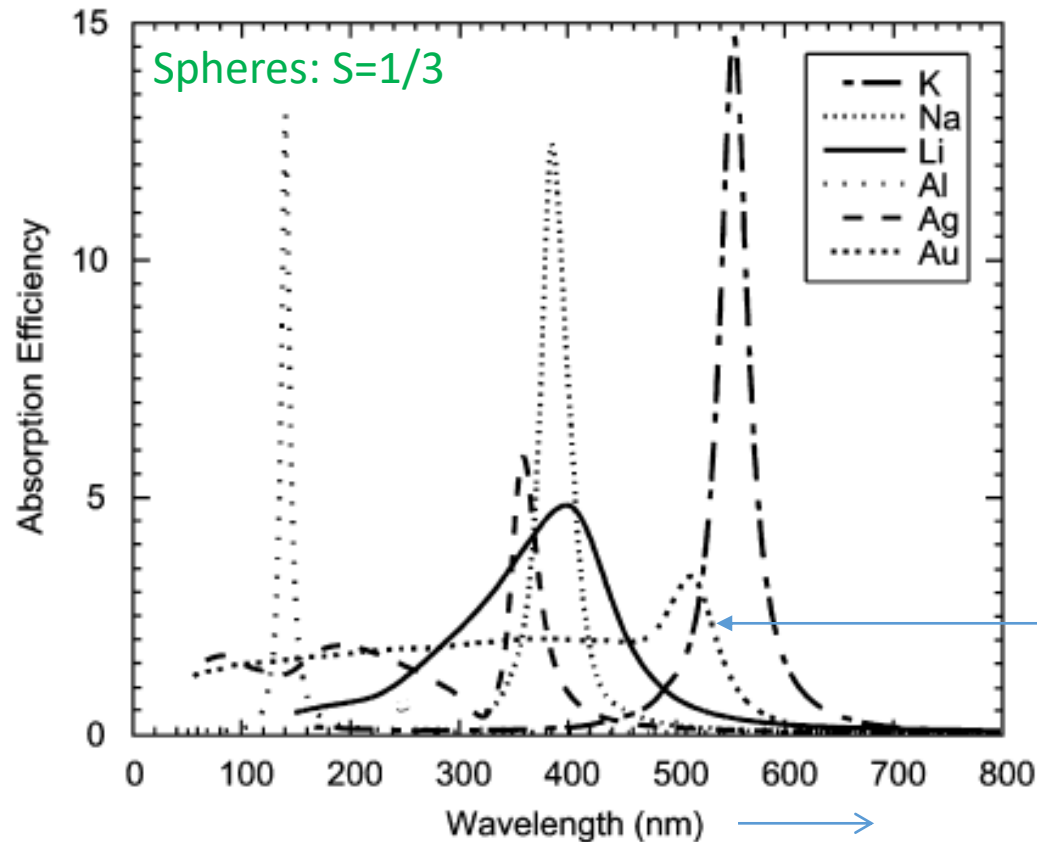
Al: $3s^2 3p^1$

Na: $3s^1$

K: $4s^1$

Lots of free e-

Big cloud



Au: $4f^{14} 5d^{10} 6s^1$

Complicated orbits

Electron scattering likely

Where is the dipole resonance of small metal spheres?

$$\alpha_{\text{sphere}} \propto \frac{1}{1/3 + 1/(\epsilon / \epsilon_b - 1)}$$

Note: unlike photonic resonance
LSP position is nearly independent of size

α max when denom near zero: $\epsilon_{\text{sphere}} / \epsilon_b \approx -2$

For Drude (inaccurate for many materials near sphere resonance): $\epsilon / \epsilon_0 \sim 1 - \frac{\omega_p^2}{\omega^2}$

$$\omega_{\text{sphere}} \sim \omega_p / \sqrt{1 + 2\epsilon_b / \epsilon_0}$$

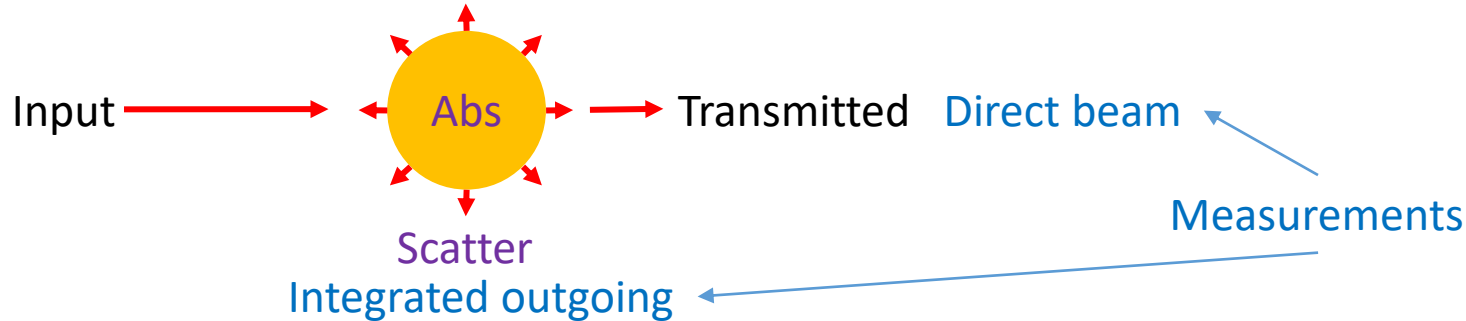
A high index background (e.g. water or glass) redshifts the resonance from free-space

In free-space: $\omega_{\text{sphere},0} \sim \omega_p / \sqrt{3}$

What's the approx. resonant wavelength of a small potassium sphere in vacuum?
How about in glass?

Far-field LSP response

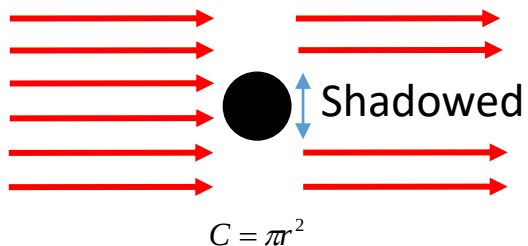
How is energy divided?



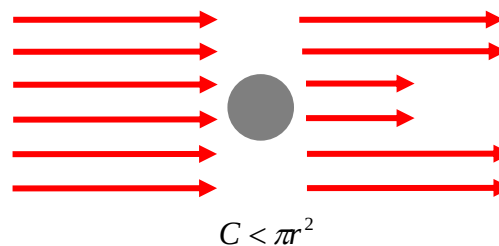
Energy balance: Transmission + Scatter + Absorption = Input
Extinction

Generally expressed as cross-sections, i.e. the equivalent “area”

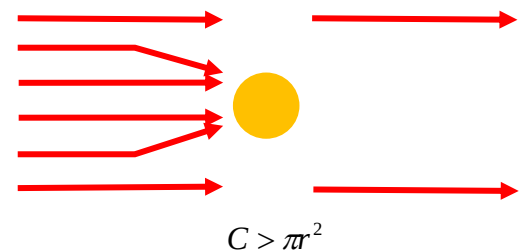
Perfect **large** absorber



Weak absorber

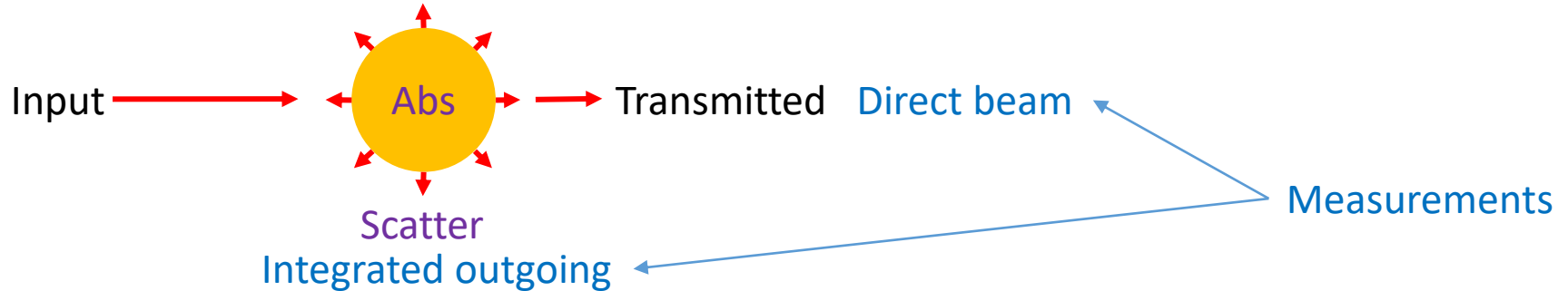


LSP can absorb **more** than area!



Far-field LSP response

How is energy divided?



Energy balance: Transmission + Scatter + Absorption = Input
Extinction

Cross-sections from (dipole) polarizability

Small particle limit:

$$C_{abs} \approx k \operatorname{Im}\{\alpha\} \quad k = 2\pi / \lambda$$

$$C_{sca} \approx \frac{\pi}{6} k^4 |\alpha|^2 \quad (\text{Rayleigh scattering})$$

Resonant cross-sections

$$\alpha_{\text{ellipsoid}} = V \frac{1}{S + 1/\chi}$$

At resonance, S cancels negative real part of $1/\chi$, leaving just the imaginary part

Drude

(inaccurate for near sphere resonance):

$$\epsilon_{\Re} - 1 \sim -\frac{\omega_p^2}{\omega^2} \quad \epsilon_I \sim \frac{\gamma \omega_p^2}{\omega^3}$$

$$\alpha_{\text{resonance}} \sim iV(\tilde{\epsilon}_{\Re} - 1)^2 / \tilde{\epsilon}_{\Im}$$

$$\alpha_{\text{resonance}} \sim iV \frac{\omega_p^2}{\gamma \omega}$$

$$\epsilon_{\Im} \ll |\epsilon_{\Re}| \quad \text{assuming low loss}$$

$$C_{\text{abs}} \approx kV(\tilde{\epsilon}_{\Re} - 1)^2 / \tilde{\epsilon}_{\Im}$$

$$C_{\text{abs}} \approx V \frac{\omega_p^2}{\gamma \omega} \quad \text{Independent of } \omega$$

$$C_{\text{sca}} \approx \frac{\pi}{6} k^4 \left[V(\tilde{\epsilon}_{\Re} - 1)^2 / \tilde{\epsilon}_{\Im} \right]^2$$

$$C_{\text{sca}} \approx \frac{\pi}{6} \left[\frac{V \omega \omega_p^2}{\gamma \omega^2} \right]^2$$

**Increases
faster with V**

Resonant cross-sections: sphere

$$\alpha_{\text{sphere}} \sim iV9/\tilde{\epsilon}_{\text{S}}$$

$$\alpha_{\text{sphere}} \sim iV \frac{\omega_p \sqrt{3}}{\gamma}$$

$$C_{\text{abs}} \approx kV9/\tilde{\epsilon}_{\text{S}}$$

$$C_{\text{abs}} \approx V \frac{\omega_p^2}{\gamma c}$$

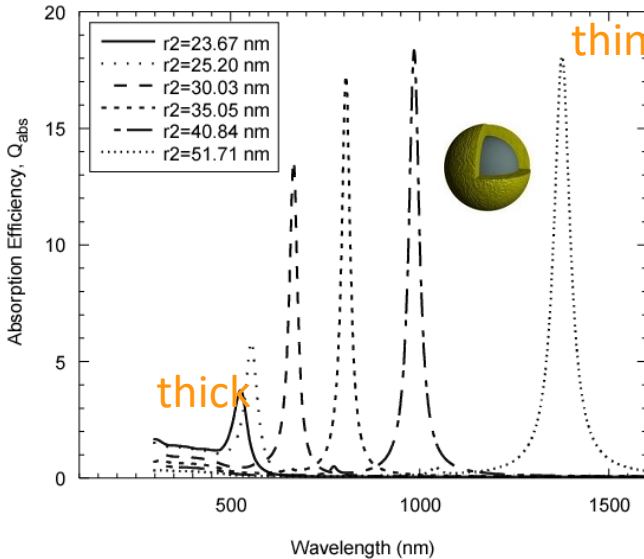
$$C_{\text{sca}} \approx \frac{\pi}{6} k^4 [V9/\tilde{\epsilon}_{\text{S}}]^2$$

$$C_{\text{sca}} \approx \frac{\pi}{6} \left[\frac{V\omega_p^3}{\gamma c^2 \sqrt{3}} \right]^2$$

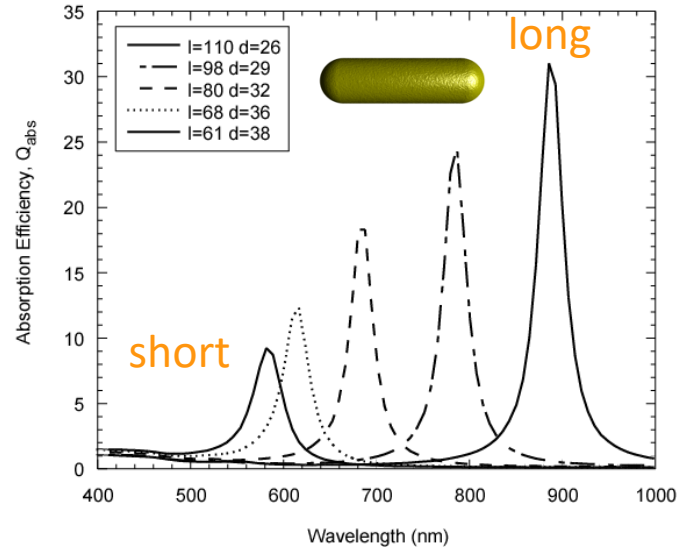
What is C_{abs} for a K sphere of 10nm radius?
Compare to the geometric cross-section.

Effect of shape on LSP

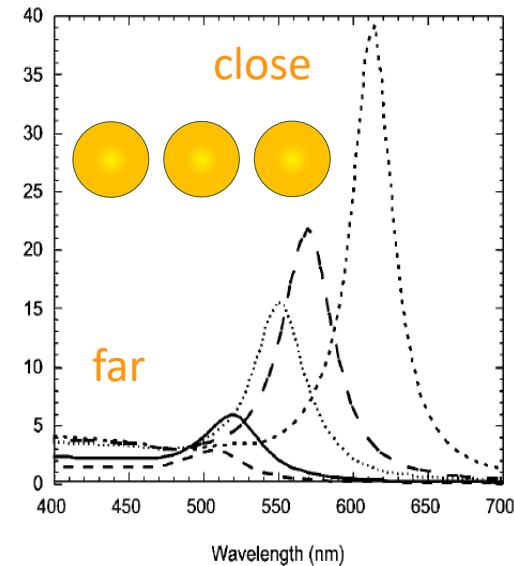
Sensitive to shape



Gold Bulletin **41** 5



$S=1/3 \longrightarrow S \rightarrow 0$



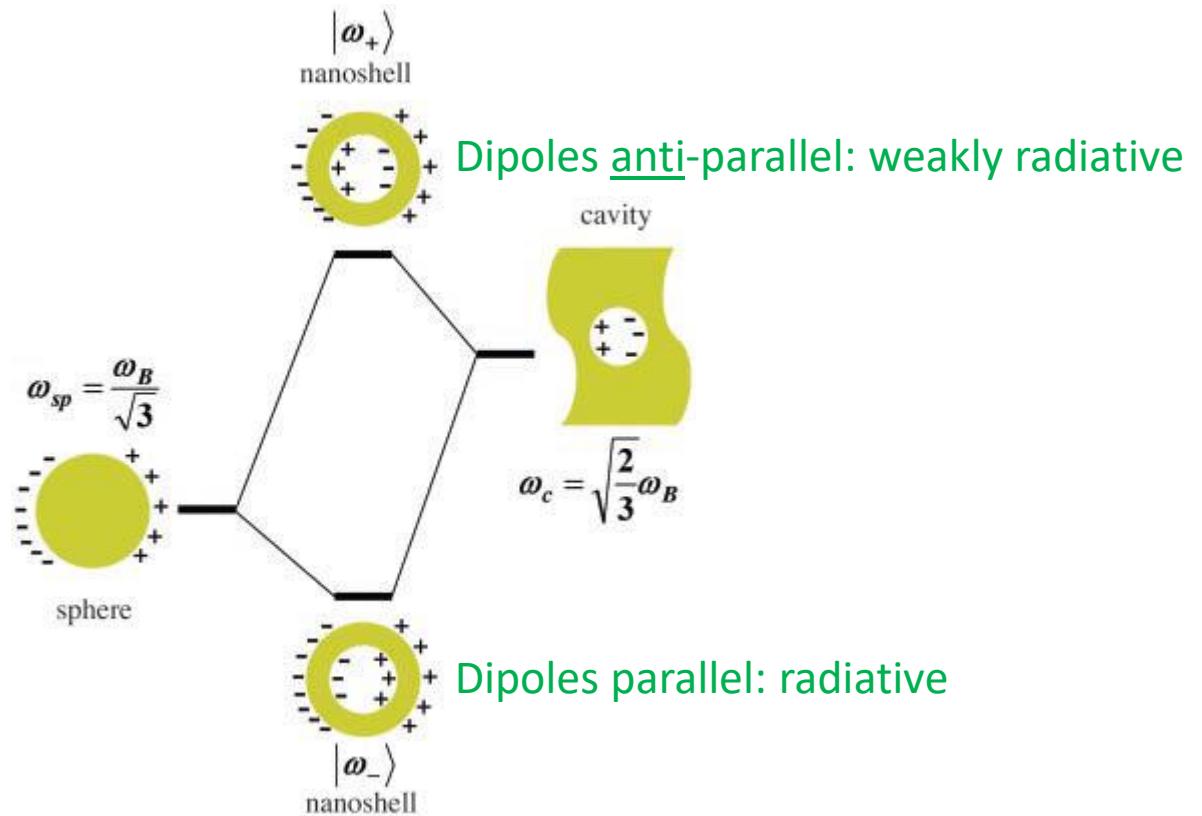
JPhysChemC **113**, 2784

Notice gets better at long λ
Primarily due to Au short λ interband absorb

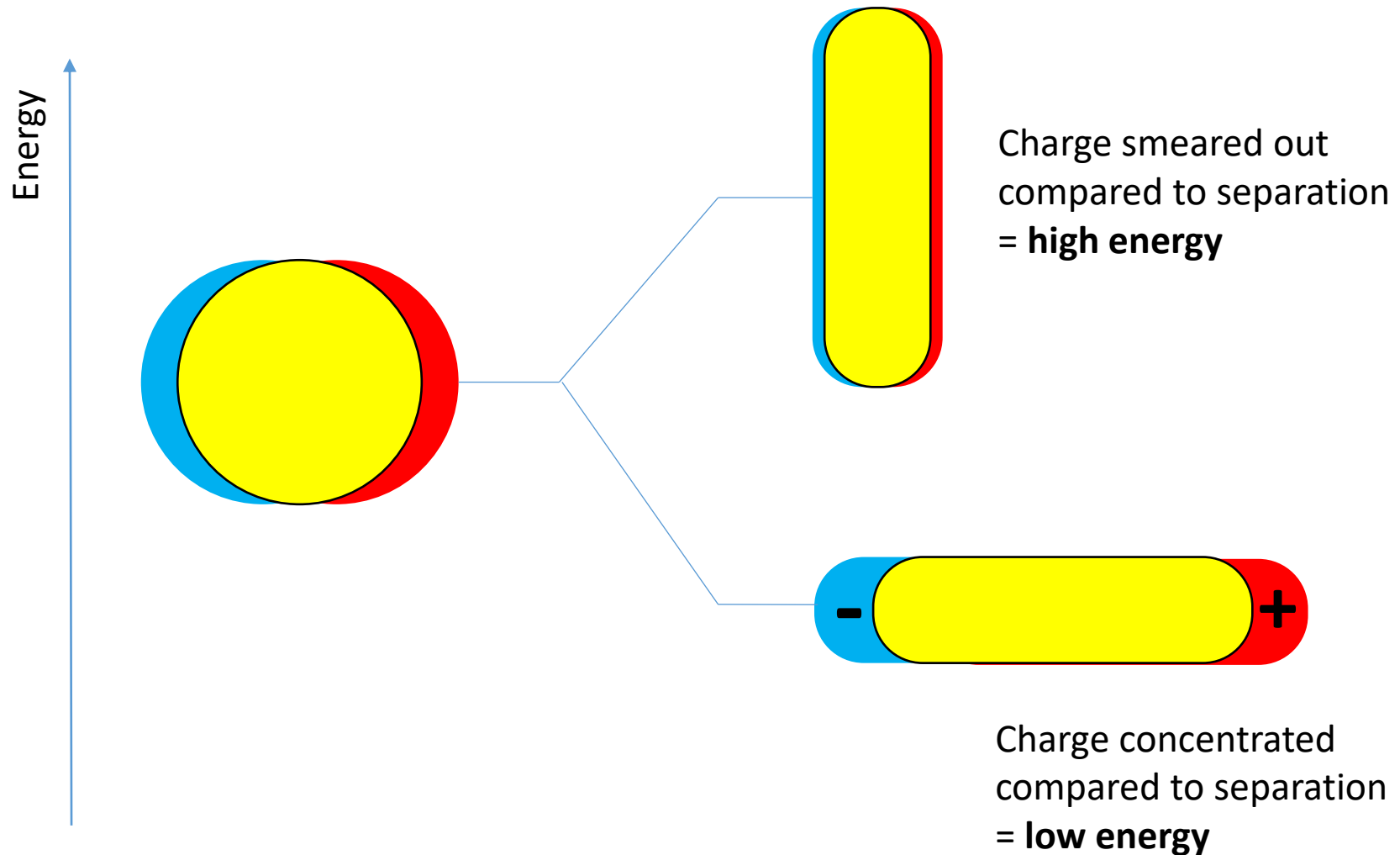
Now: let's try to understand position in terms of charge distribution

Hybridization of core-shells

Dipole modes of sphere & cavity overlap $\rightarrow$ energy splitting

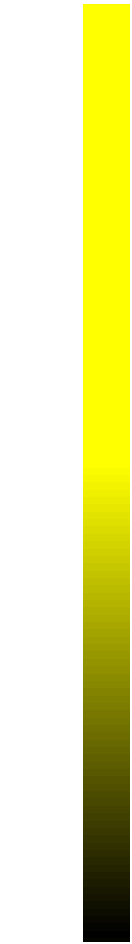


Shift of rod resonances

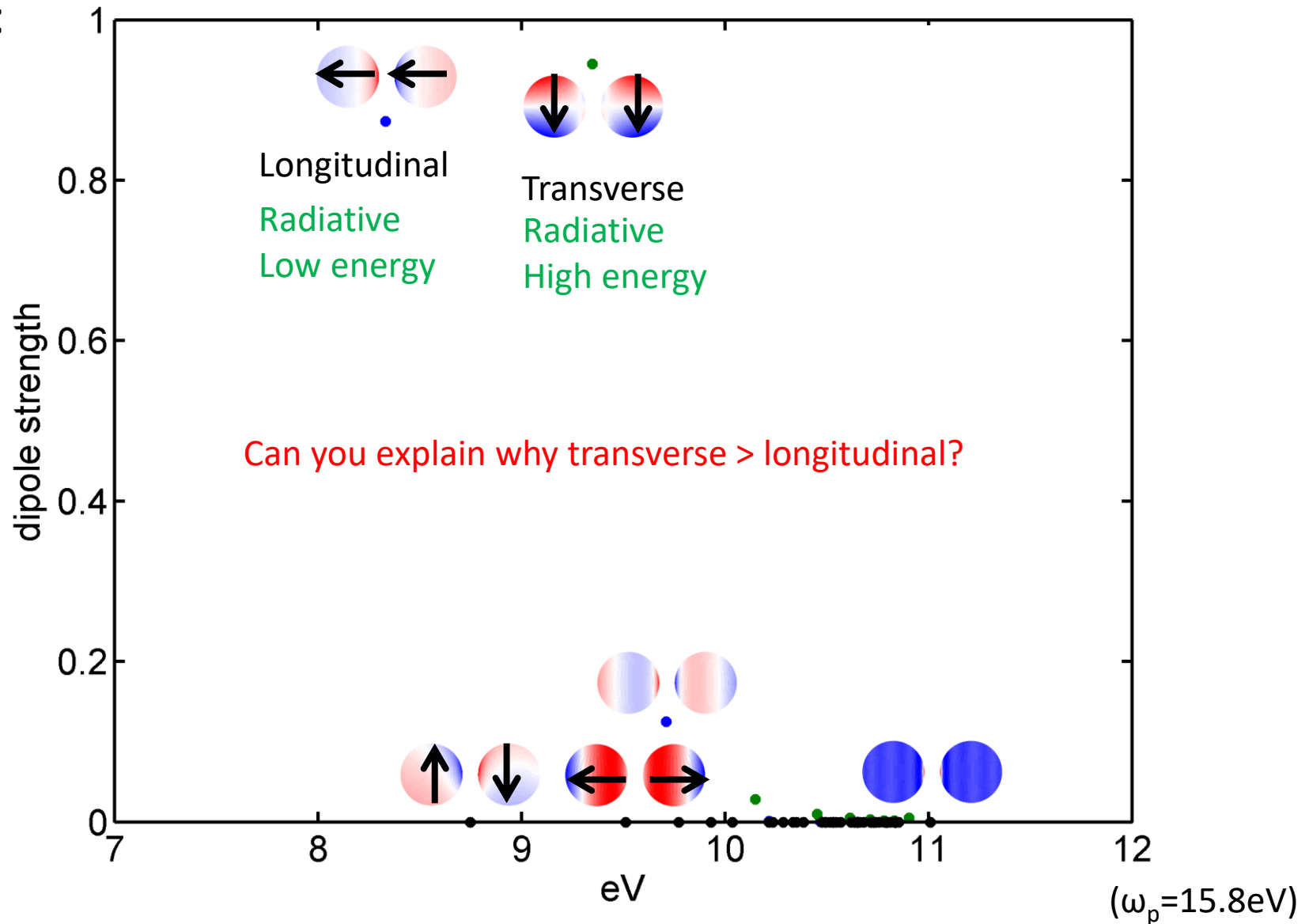


Hybridization of sphere dimer

Bright



Dark



Higher order modes

So far we have only considered dipole modes

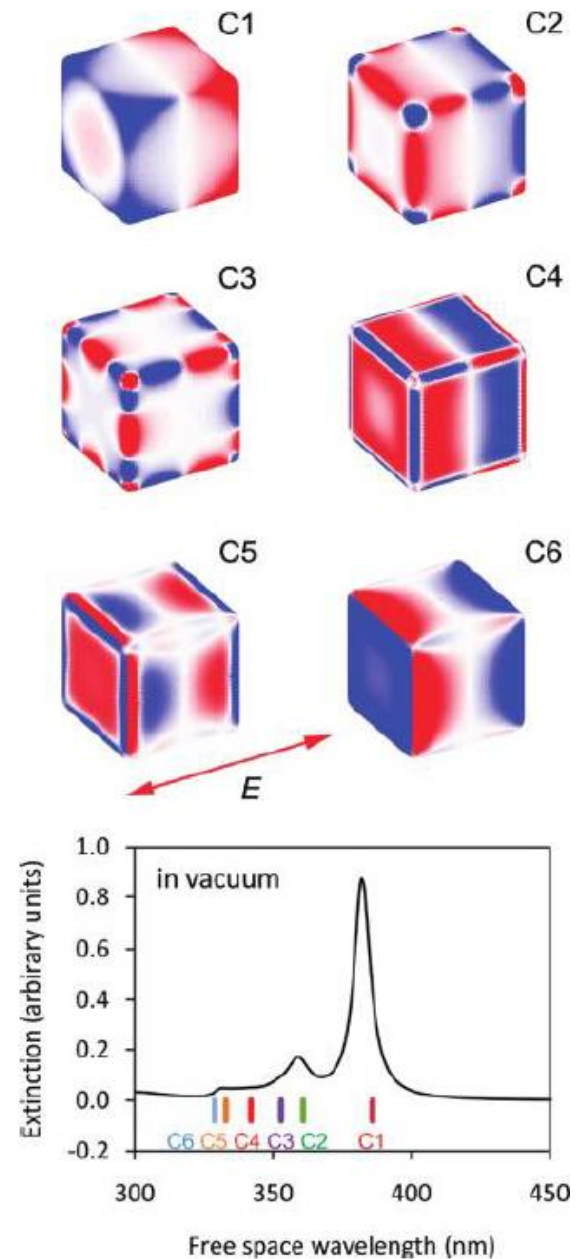
Higher order modes have more complicated distribution

- Higher energy
- Less radiative

Even ellipsoidal shapes have higher order modes

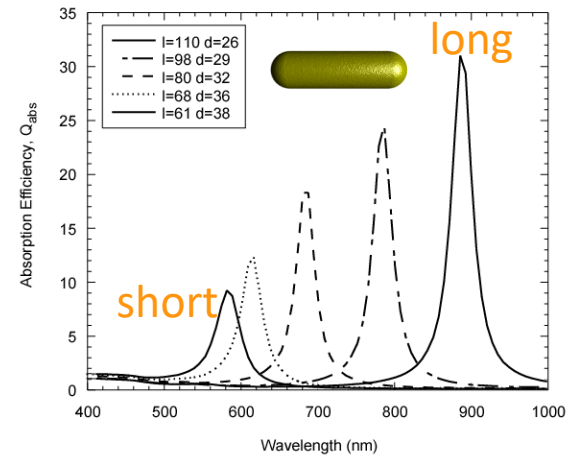
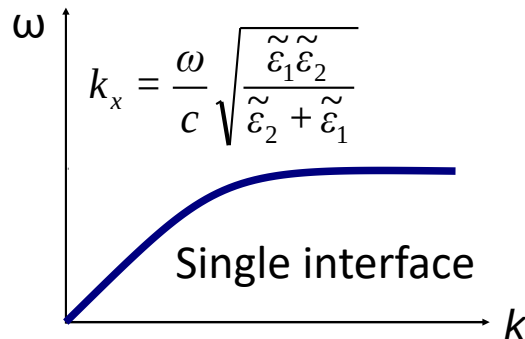
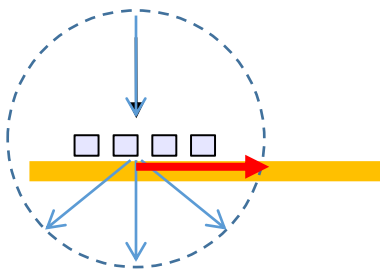
But these are not radiative until the particle becomes large

Other shapes can have very interesting modes



Summary

- Plasmons = surface resonance of conductor
- Drude model useful for basic material response
- LSP: small particle traps
 - Shape most useful for tuning
 - Electron forces key to understanding
 - Modelled by quasistatic resonance
- SPP: thin film waveguides
 - Needs special coupling (prism/gratings)
 - Highly dispersive: light-like $\rightarrow$ slow light
 - Thickness determines properties



$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi}$$

